

Tutorial - 13

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Ques 1) Evaluate the following:

$$\int_0^{\pi/4} \cos^3 2x \cdot \sin^2 4x \, dx$$

Solⁿ) $I = \int_0^{\pi/4} \sin^2 4x \cdot \cos^3 2x \cdot dx$

We know, $\sin 2\theta = 2 \sin \theta \cdot \cos \theta$

So, $\sin 4x = 2 \sin 2x \cdot \cos 2x$

$$I = \int_0^{\pi/4} (2 \sin 2x \cdot \cos 2x)^2 \cdot \cos^3 2x \, dx$$

$$= \int_0^{\pi/4} 4 \sin^2 2x \cdot \cos^2 2x \cdot \cos^3 2x \, dx$$

$$= 4 \int_0^{\pi/4} \sin^2 2x \cdot \cos^5 2x \, dx$$

Substitution,

let $2x = t$

$$\Rightarrow 2dx = dt$$

$$\Rightarrow \left[dx = \frac{dt}{2} \right]$$

limit,

x	0	$\pi/4$
t	0	$\pi/2$

limit change
0 to $\pi/2$

$$I = \int_0^{\pi/2} \sin^2 t \cdot \cos^5 t \left(\frac{1}{2} dt \right)$$

$$= \frac{1}{2} \int_0^{\pi/2} \sin^2 t \cdot \cos^5 t dt$$

Applying Walli's Formula :- $m=2, n=5$

$$= 2 \left[\frac{(2-1)(5-1)(5-3)}{(2+5)(2+5-2)(2+5-4)(2+5-6)} \right] \times 1$$

$$= 2 \left[\frac{1 \cdot 4 \cdot 2}{7 \cdot 5 \cdot 3} \right]$$

$$I = \frac{16}{105}$$

Ans!

Show that $\int_0^{\pi} x \sin^5 x \cos^4 x \, dx = \frac{8\pi}{315}$

Property :-

$$\int_0^a f(x) \, dx = \int_0^a f(a-x) \, dx$$

In this ques, $a = \pi$

So, Replace, x by $\pi - x$ in this ques.

$$I = \int_0^{\pi} (\pi - x) \sin^5(\pi - x) \cos^4(\pi - x) \, dx$$

We know, $\left[\begin{array}{l} \sin(\pi - \theta) = \sin \theta \\ \cos(\pi - \theta) = -\cos \theta \end{array} \right]^{**}$

$$I = \int_0^{\pi} (\pi - x) \sin^5 x \cdot (-\cos x)^4 \, dx$$

$$I = \int_0^{\pi} (\pi - x) \sin^5 x \cos^4 x \, dx$$

$$I = \int_0^{\pi} \pi \sin^5 x \cos^4 x - x \sin^5 x \cos^4 x \, dx$$

$$I = \int_0^{\pi} \pi \sin^5 x \cdot \cos^4 x \, dx - \int_0^a x \sin^5 x \cdot \cos^4 x \, dx$$

$$I = \pi \int_0^{\pi} \sin^5 x \cdot \cos^4 x \, dx - I$$

$$2I = \pi \int_0^{\pi} \sin^5 x \cdot \cos^4 x \, dx$$

$$m=5, \quad n=4, \quad n \Rightarrow \text{even.}$$

We know,

$$\int_0^{\pi} \sin^m x \cdot \cos^n x \, dx = 2 \int_0^{\pi/2} \sin^m x \cdot \cos^n x \, dx$$

$$2I = 2\pi \int_0^{\pi/2} \sin^5 x \cdot \cos^4 x \, dx$$

$$I = \pi \left[\frac{(5-1)(5-3)(4-1)(4-3)}{(5+4)(5+4-2)(5+4-4)(5+4-6)(5+4-8)} \right]$$

$$= \pi \left[\frac{4 \cdot 2 \cdot 3 \cdot 1}{9 \cdot 7 \cdot 5 \cdot 3 \cdot 1} \right]$$

$$I = \pi \left[\frac{24}{945} \right] \Rightarrow I = \frac{8\pi}{315} = \text{RHS}$$

Ques 3) Evaluate $\int_0^{\pi} x \sin^7 x \cdot \cos^4 x \, dx$

Solⁿ) $I = \int_0^{\pi} x \sin^7 x \cdot \cos^4 x \, dx$

Property :-

$$\int_0^a f(x) \, dx = \int_0^a f(a-x) \, dx$$

Here, $a = \pi$

So, Replace x by $(\pi-x)$ in this ques, we get,

$$I = \int_0^{\pi} (\pi-x) \sin^7(\pi-x) \cdot \cos^4(\pi-x) \, dx$$

$$= \int_0^{\pi} (\pi-x) \sin^7 x \cdot \cos^4 x \cdot dx$$

$$= \int_0^{\pi} \pi \sin^7 x \cdot \cos^4 x \, dx - \int_0^{\pi} x \sin^7 x \cdot \cos^4 x \, dx$$

$$I = \int_0^{\pi} \pi \sin^7 x \cdot \cos^4 x \, dx - I$$

$$\Rightarrow 2I = \pi \int_0^{\pi} \sin^7 x \cdot \cos^4 x \, dx$$

$$\Rightarrow \cancel{2I} = \cancel{2\pi} \int_0^{\pi/2} \sin^7 x \cdot \cos^4 x \, dx$$

$$\Rightarrow I = \pi \int_0^{\pi/2} \sin^7 x \cdot \cos^4 x \, dx$$

$$I = \pi \left[\frac{(7-1)(7-3)(7-5)(4-1)(4-3)}{(7+4)(7+4-2)(7+4-4)(7+4-6)(7+4-8)(7+4-10)} \right]$$

$$I = \pi \left[\frac{6 \cdot 4 \cdot 2 \cdot 3 \cdot 1}{11 \cdot 9 \cdot 7 \cdot 5 \cdot 3 \cdot 1} \right]$$

$$\cancel{\pi} \left[\frac{\cancel{14} \cancel{4}}{\cancel{20} \cancel{7} \cancel{9} \cancel{0}} \right] = \pi \left[\frac{144}{10395} \right]$$

$$I = \frac{16\pi}{1155}$$

Ans!

(Ques 4) If $I_n = \int_{\pi/4}^{\pi/2} \cot^n \theta \, d\theta$, prove that

$$I_n = \frac{1}{n-1} - I_{n-2} \text{ . Hence evaluate}$$

$$\int_{\pi/4}^{\pi/2} \cot^6 \theta \, d\theta.$$

(Solⁿ) $I_n = \int_{\pi/4}^{\pi/2} \cot^n \theta \, d\theta$

$$= \int_{\pi/4}^{\pi/2} \cot^{n-2} \theta \cdot \cot^2 \theta \, d\theta$$

$$= \int_{\pi/4}^{\pi/2} \cot^{n-2} \theta \cdot (\operatorname{cosec}^2 \theta - 1) \, d\theta$$

$$I_n = \int_{\pi/4}^{\pi/2} \cot^{n-2} \theta \cdot \operatorname{cosec}^2 \theta \, d\theta - \underbrace{\int_{\pi/4}^{\pi/2} \cot^{n-2} \theta \, d\theta}_{I_{n-2}}$$

$$I_n = \int_{\pi/4}^{\pi/2} \cot^{n-2} \theta \cdot \operatorname{cosec}^2 \theta \, d\theta - I_{n-2}$$

let, $t = \cot \theta$
 $dt = -\operatorname{cosec}^2 \theta \cdot d\theta \rightarrow d\theta = \frac{-dt}{\operatorname{cosec}^2 \theta}$

limit,

θ	$\pi/4$	$\pi/2$
t	1	0

$$I_n = \int_1^0 t^{n-2} \cdot \cancel{\operatorname{cosec}^2 \theta} \cdot \left(\frac{-dt}{\cancel{\operatorname{cosec}^2 \theta}} \right) - I_{n-2}$$

$$= \int_0^1 t^{n-2} \, dt - I_{n-2}$$

$$= \left[\frac{t^{n-1}}{n-1} \right]_0^1 - I_{n-2}$$

$$= \frac{1^{n-1}}{n-1} - \frac{0^{n-1}}{n-1} - I_{n-2}$$

$$= \frac{1}{n-1} - 0 - I_{n-2} \quad (\text{Assuming } n > 1)$$

$$I_n = \frac{1}{n-1} - I_{n-2}$$

$\text{Proved} \quad \textcircled{i}$

Evaluating $\int_{\pi/4}^{\pi/2} \cot^6 \theta \, d\theta$

In eqⁿ (i) put $n=6$,

$$I_6 = \frac{1}{6-1} - I_{6-2}$$

$$I_6 = \frac{1}{5} - I_4$$

$$= \frac{1}{5} - \left[\frac{1}{3} - I_2 \right]$$

$$= \frac{1}{5} - \frac{1}{3} + I_2$$

$$= \frac{1}{5} - \frac{1}{3} + \left[1 - I_0 \right]$$

$$= \frac{1}{5} - \frac{1}{3} + 1 - \frac{\pi}{4}$$

$$= \frac{3-5+15}{15} - \frac{\pi}{4}$$

$$I_6 = \frac{13}{15} - \frac{\pi}{4}$$

Ans!

$$\# I_0 = \int_{\pi/4}^{\pi/2} \cot^0 \theta \, d\theta$$

$$= \frac{\pi}{2} - \frac{\pi}{4}$$

$$I_0 = \frac{\pi}{4}$$

Ques 5) if $I_n = \int_0^{\pi/4} \frac{\sin(2n-1)x}{\sin x} dx$, then prove that

$$n(I_{n+1} - I_n) = \sin \frac{n\pi}{2} \text{ and hence}$$

find I_3 .

Solⁿ) Taking,

$$I_{n+1} - I_n = \int_0^{\pi/4} \frac{\sin(2(n+1)-1)x}{\sin x} dx -$$

$$\int_0^{\pi/4} \frac{\sin(2n-1)x}{\sin x} dx$$

$$= \int_0^{\pi/4} \frac{\sin(2n+1)x - \sin(2n-1)x}{\sin x} dx$$

We know, $\sin C - \sin D = 2 \cos \frac{C+D}{2} \cdot \sin \frac{C-D}{2}$
Applying,

$$= \int_0^{\pi/4} \frac{2 \cos 2nx \cdot \sin x}{\sin x} dx$$

$$= 2 \int_0^{\pi/4} \cos 2nx \, dx$$

$$= \cancel{2} \left[\frac{\sin 2nx}{\cancel{2}n} \right]_0^{\pi/4}$$

$$= \left[\frac{\sin 2nx}{n} \right]_0^{\pi/4}$$

$$= \frac{\sin 2n \left(\frac{\pi}{4} \right)}{n} - \frac{\sin 2 \cdot n \cdot 0}{n}$$

$$= \frac{\sin \left(\frac{n\pi}{2} \right)}{n} - 0$$

$$\Rightarrow I_{n+1} - I_n = \frac{1}{n} \left[\sin \frac{n\pi}{2} \right]$$

$$\Rightarrow \boxed{n(I_{n+1} - I_n) = \sin \frac{n\pi}{2}} \quad \text{--- (i)}$$

Hence proved.

$$I_3 = ? \quad \text{P.T.O}$$

$$I_3 = ?$$

put, $n = 2$ in eqⁿ ①

we get,

$$2(I_3 - I_2) = \sin \frac{2\pi}{2}$$

$$I_3 - I_2 = 0 \times \frac{1}{2} = 0$$

$$\boxed{I_3 = I_2}$$

put $n = 1$ in eqⁿ ① we get,

$$\Rightarrow 1(I_2 - I_1) = \sin \pi/2$$

$$\Rightarrow I_2 - I_1 = 1$$

$$\Rightarrow \boxed{I_2 = 1 + I_1}$$

$$I_1 = ?$$

$$\text{put } n = 1 \text{ in } I_n = \int_0^{\pi/4} \frac{\sin(2n-1)x}{\sin x} dx$$

$$I_1 = \int_0^{\pi/4} dx = [x]_0^{\pi/4} = \pi/4 - 0$$

$$I_1 = \frac{\pi}{4}$$

So, $I_2 = 1 + \frac{\pi}{4}$

$$I_2 = I_3 = 1 + \frac{\pi}{4}$$

Ans!
=

Ques 6) If $I_n = \int_0^{\pi/4} \sec^n x dx$, then prove that I_n ,

$$I_n = \frac{(\sqrt{2})^{n-2}}{n-1} + \left(\frac{n-2}{n-1} \right) I_{n-2} \text{ and hence}$$

evaluate $\int_0^{\pi/4} \sec^6 x dx$

Solⁿ $I_n = \int_0^{\pi/4} \sec^{n-2} x \cdot \sec^2 x dx$

let, $u = \sec^{n-2} x$

$$\Rightarrow du = (n-2) \sec^{n-3} x (\sec x \tan x) dx$$

$$\Rightarrow du = (n-2) \sec^{n-2} x \cdot \tan x dx$$

let,

$$v = \tan x$$

$$dv = \sec^2 x dx$$

Using $\int u dv = uv - \int v du$

$$I_n = \left[\sec^{n-2} x \tan x \right]_0^{\pi/4} - \int_0^{\pi/4} \tan x \cdot (n-2) \sec^{n-2} x \cdot \tan x dx$$

Evaluating the limit :-

$$\left[\sec^{n-2} x \cdot \tan x \right]_0^{\pi/4} = \sec^{\frac{n-2}{4}} \cdot \tan \frac{\pi}{4} -$$

$$\sec^{n-2}(0) \cdot \tan(0)$$

$$= (\sqrt{2})^{n-2} \cdot 1 - 0$$

$$= (\sqrt{2})^{n-2}$$

Simplifying integral term :-

$$-(n-2) \int_0^{\pi/4} \sec^{n-2} x (\sec^2 x - 1) dx$$

$$= -(n-2) \int_0^{\pi/4} (\sec^n x - \sec^{n-2} x) dx$$

$$= -(n-2) \int_0^{\pi/4} \sec^n x dx + (n-2) \int_0^{\pi/4} \sec^{n-2} x dx$$

$$= -(n-2) I_n + (n-2) I_{n-2}$$

Combining the terms :-

$$\Rightarrow I_n = (\sqrt{2})^{n-2} - (n-2) I_n + (n-2) I_{n-2}$$

$$\Rightarrow I_n + (n-2) I_n = (\sqrt{2})^{n-2} + (n-2) I_{n-2}$$

$$\Rightarrow (n-1) I_n = (\sqrt{2})^{n-2} + (n-2) I_{n-2}$$

$$I_n = \frac{(\sqrt{2})^{n-2}}{n-1} + \left(\frac{n-2}{n-1}\right) I_{n-2}$$

—(i)

Hence
proved.

$$I_6 = ?$$

For I_6 , put $n=6$ in eqⁿ (i)

$$I_6 = \frac{(\sqrt{2})^{6-2}}{6-1} + \left(\frac{6-2}{6-1}\right) I_{6-2}$$

$$= \frac{4}{5} + \frac{4}{5} I_4$$

$$= \frac{4}{5} + \frac{4}{5} \left[\frac{2}{3} + \frac{2}{3} I_2 \right]$$

$$= \frac{4}{5} + \frac{4}{5} \left[\frac{2}{3} + \frac{2}{3} \cdot 1 \right]$$

$$= \frac{4}{5} + \frac{4}{5} \left(\frac{4}{3} \right)$$

$$I_6 = \frac{28}{15}$$

Ans!

Tutorial - 14

Ques 1) Evaluate $\int_0^{\infty} x^7 \cdot e^{-2x^2} dx$

Solⁿ) $I = \int_0^{\infty} x^7 \cdot e^{-2x^2} dx$

let, $2x^2 = t$

$\Rightarrow 4x dx = dt$

$\Rightarrow dx = \frac{dt}{4x}$

$dx = \frac{dt}{4\left(\sqrt{\frac{t}{2}}\right)}$

 $2x^2 = t$
 $x^2 = \frac{t}{2}$
 $(x = \sqrt{\frac{t}{2}})$

limit,

x	0	∞
t	0	∞

$\therefore I = \int_0^{\infty} \left(\sqrt{\frac{t}{2}}\right)^7 \cdot e^{-t} \cdot \frac{dt}{4\left(\sqrt{\frac{t}{2}}\right)}$

$$\Rightarrow I = \int_0^{\infty} \left(\frac{t}{2}\right)^{7/2} \cdot e^{-t} \cdot \frac{dt}{4 \left(\frac{t}{2}\right)^{1/2}}$$

$$= \int_0^{\infty} \frac{e^{-t} \cdot t^{7/2} \cdot t^{-1/2} dt}{2^{7/2} \cdot 2^{-1/2} \cdot 4}$$

$$= \int_0^{\infty} \frac{e^{-t} \cdot t^{7/2 - 1/2} dt}{2^{7/2 - 1/2} \cdot 4}$$

$$= \int_0^{\infty} \frac{e^{-t} \cdot t^3 dt}{2^3 \cdot 4}$$

$$= \frac{1}{32} \int_0^{\infty} e^{-t} \cdot t^3 dt$$

$$= \frac{1}{32} \int_0^{\infty} e^{-t} \cdot t^{4-1} dt$$

$$= \frac{1}{32} \cdot \sqrt{4}$$

$$* \sqrt{4} = \sqrt{3+1} = \underline{2}$$

$$I = \frac{1}{32} \cdot 3!$$

$$\# 3! = 6$$

$$I = \frac{6}{32}$$

Ans!

Ques 2) Evaluate $\int_0^1 (x \log x)^3 dx$

Solⁿ) $I = \int_0^1 (x \log x)^3 dx$

let, $\log x = -t$

Rewrite $\rightarrow e^{-t} = x$

differentiation $\rightarrow -e^{-t} dt = dx$

limit,

x	0	1
t	∞	0

$$\# \left\{ \begin{array}{l} \log x = -t \\ \log 1 = 0 \\ \log 0 = -\infty \\ \text{So, } -t = -\infty \\ (t = \infty) \end{array} \right.$$

$$I = \int_{\infty}^0 (e^{-t} \cdot (-t))^3 \cdot (-e^{-t} dt)$$

$$= \int_0^{\infty} e^{-3t} \cdot (-t)^3 \cdot e^{-t} \cdot dt$$

$$I = - \int_0^{\infty} e^{-4t} \cdot t^3 \cdot dt$$

we know, $\int_0^{\infty} e^{-Kx} \cdot x^{n-1} dx = \frac{\Gamma n}{K^n}$

(P-6)

So,

$$- \int_0^{\infty} e^{-4t} \cdot t^{4-1} \cdot dt = - \frac{\Gamma 4}{4^4}$$

$$I = \left(- \frac{3!}{256} \right) \text{Ans!}$$

Ques 3) Show that $\int_0^{\infty} 3^{-4x^2} dx = \frac{\sqrt{\pi}}{4\sqrt{\log 3}}$

Solⁿ) $I = \int_0^{\infty} 3^{-4x^2} \cdot dx$

let,

$$3^{4x^2} = e^t$$

So,

$$\Rightarrow 4x^2 \cdot \log 3 = t$$

$$\Rightarrow x^2 = \frac{t}{4 \log 3} \Rightarrow x = \frac{\sqrt{t}}{\sqrt{4 \log 3}}$$

$$\Rightarrow dx = \frac{1}{2\sqrt{x}} \cdot \frac{1}{\sqrt{4\log 3}} \cdot dt$$

limit,

x	0	∞
t	0	∞

$$\therefore I = \int_0^{\infty} e^{-t} \cdot \frac{1}{2\sqrt{t} \cdot \sqrt{4\log 3}} dt$$

$$= \int_0^{\infty} e^{-t} \cdot \frac{1}{2\sqrt{t}} \cdot \frac{1}{2\sqrt{\log 3}} dt$$

$$= \frac{1}{4\sqrt{\log 3}} \int_0^{\infty} e^{-t} \cdot t^{-1/2} dt$$

$$= \frac{1}{4\sqrt{\log 3}} \cdot \int_0^{\infty} e^{-t} \cdot t^{1/2-1} dt$$

So,

$$= \frac{1}{4\sqrt{\log 3}} \cdot \Gamma_{1/2}$$

$$\left(\sqrt{\frac{1}{2}} - \sqrt{\pi} \right)$$

$$I = \frac{\sqrt{\pi}}{4\sqrt{\log 3}} = \text{RHS} \rightarrow \text{Hence Proved.}$$

Ques 4) Evaluate $\int_0^1 x^3 (1 - \sqrt{x})^5 dx$

Solⁿ) Basic definition of Beta :-

$$\beta(m, n) = \int_0^1 x^{m-1} (1-x)^{n-1} dx$$

Now, put $\sqrt{x} = t$

$$\Rightarrow \frac{1}{2\sqrt{x}} dx = dt$$

$$\Rightarrow dx = 2\sqrt{x} dt$$

$$\Rightarrow \boxed{dx = 2t dt}$$

Limit,

x	0	1
t	0	1

$$\therefore I = \int_0^1 (t^2)^3 \cdot (1-t)^5 \cdot 2t dt$$

$$= \int_0^1 t^6 \cdot (1-t)^5 \cdot 2t \cdot dt$$

$$= 2 \int_0^1 t^7 (1-t)^5 dt$$

$$I = \frac{1}{2} \int_0^1 t^{8-1} (1-t)^{6-1} dt$$

$$\# m=8, n=6$$

$$I = 2 \beta(8, 6)$$

$$I = 2 \frac{8! 6!}{(8+6)!}$$

$$I = \frac{2 \cdot 7! \cdot 5!}{13!}$$

Ans.

OR

$$I = \frac{1}{5148}$$

→ Simplify. Ans!

Ques 5) Evaluate $\int_0^{\infty} \frac{x^8 (1-x^6)}{(1+x)^{24}} dx$

Soln) $I = \int_0^{\infty} \frac{x^8 (1-x^6)}{(1+x)^{24}} dx$

$$= \int_0^{\infty} \frac{x^8 - x^{14}}{(1+x)^{24}} dx$$

$$= \int_0^{\infty} \frac{x^8}{(1+x)^{24}} dx - \int_0^{\infty} \frac{x^{14}}{(1+x)^{24}} dx$$

$$= \beta(9, 15) - \beta(15, 9)$$

$$= 0$$

{ By property No. 1
i.e.

$$\beta(m, n) = \beta(n, m) }$$

So, $I = 0$

Ans!

Ques 6) Evaluate $\int_2^5 (x-2)^3 (5-x)^2 dx$

Solⁿ) $I = \int_2^5 (x-2)^3 (5-x)^2 dx$

let,

$x-2 = (5-2)t$

$x = 2 + (5-2)t$

$\boxed{x = 2 + 3t}$ diff. $\rightarrow dx = 3 dt$

limit,

x	2	5
t	0	1

$\therefore I = \int_0^1 (3t)^3 [5 - (3t + 2)]^2 \cdot 3 dt$

$= \int_0^1 3^3 \cdot t^3 [3 - 3t]^2 \cdot 3 dt$

$= \int_0^1 3^4 \cdot t^3 \cdot 3^2 \cdot (1-t)^2 dt$

$= 3^6 \int_0^1 t^3 (1-t)^2 dt$

$$I = 3^6 \int_0^1 t^{4-1} (1-t)^{3-1} dt$$

$$= 3^6 \beta(4, 3)$$

$$= 3^6 \cdot \frac{14 \cdot 13}{\sqrt{4+3}}$$

$$I = 3^6 \cdot \frac{3! \cdot 2!}{6!}$$

Ans!